12-04 第2篇

阅读理解（说明文） Why Do Bees "Dance" to Tell Each Other Where the Flowers Are?

① On a hot June afternoon in 1944, an Austrian scientist named **Karl von Frisch** was watching honeybees inside a thin glass-walled hive. He had placed a small dish of sugar-water **60 metres** east of the hive, under a walnut tree. Within a few minutes, one bee found the dish, filled her honey stomach, and flew home. What happened next was so strange that von Frisch rubbed his eyes. **Inside the hive, on a vertical comb, the returning bee began to run in a small figure-of-eight shape, wagging her abdomen sharply as she ran along the middle line.** Other bees crowded close, felt her with their antennae — and within two minutes, **fifteen of them** flew out of the hive, straight to the east, and landed on the sugar-water dish. The bee, von Frisch realised, had just **told them where the food was**. She had done it by **dancing**. For this discovery, honeybees would one day share, with humans, the 1973 Nobel Prize in Physiology or Medicine — and turn out to be among the most surprising communicators on Earth.

② The "dance" that von Frisch watched is called, today, the **waggle dance**. It is performed only when the food source is **more than about 80 metres** from the hive. (For closer food, bees do a simpler "round dance" — they just run in a circle, and other bees go out and search nearby.) The waggle dance is a figure-of-eight with a **straight middle line**, along which the bee wiggles her body from side to side, about **13 waggles per second**. She repeats this figure five, ten, sometimes thirty times in a row. On the outside, she looks a little mad — a small golden animal jogging in a strange, tight little pattern. But every single movement, it turns out, is **carrying information**.

③ **The direction of the straight middle line codes the direction of the flowers.** Here is the most surprising part. Inside the hive, on a vertical comb, the bee cannot see the sun. But she treats **"straight up"** on the comb as **"toward the sun"** in the world outside. If the flowers are exactly toward the sun, her waggle line points straight up. If the flowers are, say, **30° to the right of the sun**, her waggle line tilts **30° to the right of vertical**. If they are directly away from the sun, she waggles **straight down**. This is a beautiful little translation from three-dimensional sunlight into two-dimensional comb. Her sisters, feeling her on the dark comb, translate back — and fly out in the right direction, **accurate to about 7—10°** over distances of hundreds of metres.

④ **The duration of the straight middle line codes the distance.** The longer the bee waggles, the farther the flowers are. On average, about **one second of waggling corresponds to about 1 kilometre** away from the hive. A tiny patch of flowers at 200 metres: about 0.2 seconds of waggle per figure. A rich patch at 3 km: three seconds. A 2019 study at the University of Sussex, which recorded more than 2,500 waggle dances

with high-speed cameras, found the accuracy was remarkable: bees that followed the dance found the food source **within 30 metres** of the true spot in 84% of cases, even at 3 km distance. That is like telling a friend, by a short dance, to find a specific tree in a park **several hundred paces away**.

⑤ **The dance also tells others how good the food is.** A rich, sugary flower source makes the bee **repeat the dance more times** (sometimes a hundred figures in a row) and **wiggle more violently**. A thin, so-so source might get only three or four polite figures. Experienced watcher-bees in the hive sense this "excitement" and adjust how many of them set out. A 2022 German study found that a dance judged "high excitement" recruited **about five times** as many new foragers as a "low excitement" dance. The hive, essentially, runs a quiet, careful vote: better food, more voters, faster response.

⑥ **What about on cloudy days when bees cannot see the sun?** A 1949 experiment by von Frisch's team gave us the answer: bees can still read the **polarisation pattern** of sunlight through clouds — the invisible structure of scattered sky-light that keeps the "direction of the sun" readable even behind grey weather. As long as a **small patch of clear sky** is visible, the waggle dance still points correctly. In heavy fog, though, bees stop dancing and just rest. They are, reasonably, honest about their limits.

⑦ And what if the food is found **after sunset, when the sun has gone?** Here the bee does something almost philosophical. She remembers where the sun was **at the time she found the food**, and then **updates her dance angle continuously** as time passes, using her internal clock and her sense of how the sun moves across the sky. A 1956 study showed that a bee forced to stay in the hive for three hours after finding food would, when she finally danced, use the **correct updated sun position** — not the old one. Inside that small golden head is a quiet little astronomer.

⑧ Why did evolution invent all this? **Because honeybee food is scattered, brief, and far.** A patch of wildflowers might open for only two days, 2 km away, in one small corner of a meadow. If a single bee found it and kept the secret, the hive could only send one bee at a time — too slow for the short bloom. By **sharing the map** through dance, one bee can send **dozens** of sisters in the right direction within minutes. It is a species-level cooperative design. Humans have maps; bees have dances. The truth they both express is the same: *"go that way, for this long, it is worth it."*

⑨ **A quiet final thought.** For 80 million years before humans, bees had already invented a kind of short, careful language — with direction, distance, and enthusiasm — all packed into a small dance on a dark comb. The next time you see a bee quickly leave a clover flower, imagine her flying home and doing a small, precise dance for her sisters. Inside that one-second waggle, there is a **kilometre of sunlight**, a **patch of spring flowers**, and, quietly, **the oldest map in the world**.

注释： hive 蜂巢；comb 蜂巢板；waggle 左右摆动；forager 觅食蜂；polarisation 偏振；abdomen 腹部

根据短文内容，选择最佳答案。

1. What did Karl von Frisch observe in 1944 that surprised him?

A. A bee drank all the sugar-water by herself.
B. A returning bee, back inside the hive on a vertical comb, ran in a small figure-of-eight shape, wagging her abdomen along the middle line; within two minutes, fifteen other bees flew out and landed exactly on the sugar-water dish 60 metres east — she had told them where the food was.
C. Bees attacked the walnut tree.
D. Bees killed one another.

2. According to Paragraph ③, what does the direction of the straight middle line of the waggle dance code?

A. The colour of the flowers.
B. The smell of the flowers.
C. The direction of the flowers, translated onto the vertical comb: "straight up" means toward the sun, and a 30° tilt right of vertical means 30° to the right of the sun; sisters on the comb re-translate this into real flight direction, accurate to about 7—10°.
D. The shape of the flowers.

3. What does the duration of the middle line code, and what did the 2019 Sussex study find?

A. It codes the temperature; the study found bees prefer warm days.
B. It codes the distance — roughly one second of waggling corresponds to about 1 km; the Sussex study recorded over 2,500 dances and found that bees following the dance landed within 30 metres of the true spot in 84% of cases, even at 3 km distance.
C. It codes the food's colour; the study found bees prefer yellow.
D. It codes the number of flowers exactly.

4. According to Paragraph ⑦, what does a bee do if she has to wait in the hive for three hours after finding the food before she can dance?

A. She dances with the old sun position.
B. She refuses to dance.
C. She updates the dance angle continuously using her internal clock and her sense of the sun's movement, so that when she finally dances, she uses the correct updated sun position — not the old one.
D. She dances the wrong way.

5. What is the main point of the passage?

A. Bees cannot communicate at all.
B. Over 80 million years, honeybees have evolved a surprisingly precise "language" — the

waggle dance — that packs direction (via the line's angle on the vertical comb, re-translated to the sun), distance (via the duration of waggling), and quality (via the number of repetitions and intensity) into a small dance on a dark comb, making it possible for one scout bee to send many sisters quickly to scattered, short-lived flower patches.
C. Bees only dance when humans watch them.
D. The waggle dance is a kind of attack signal.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第①段 1944 年观察
【解析】"8 字形 / 15 只 / 60 米东"直引。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段方向编码
【解析】"向上 = 向日 + 30° 倾斜"直引。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段距离编码 + 苏塞克斯
【解析】"1 秒 ≈ 1 km / 84%"直引。A、C、D 均错。故选 B。

4.
【答案】C
【题型】细节理解题
【定位】第⑦段"内部时钟"
【解析】"持续更新角度"直引。A、B、D 均错。故选 C。

5.
【答案】B
【题型】主旨大意题
【定位】全文五段总括
【解析】B 对方向 + 距离 + 质量三维总结。A、C、D 均偏离。故选 B。